Confusion Matrix, ROC_AUC and Imbalanced Classes in Logistic Regression

Lily Su · Follow

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To review basic underlying concepts, precision is the measure of how out of all your positive predictions, how many were correct. Recall is out of all the times you predicted positive how many total actually in the sample were positive (including the ones you missed).

Generating a Confusion Matrix:

```
from sklearn.metrics import classification_report, confusion_matrix

threshold = 0.1
y_pred = y_pred_proba >= threshold

print(classification_report(y_train, y_pred))

pd.DataFrame(confusion_matrix(y_train, y_pred),
             columns=['Predicted Negative', 'Predicted Positive'],
             index=['Actual Negative', 'Actual Positive'])
```

	precision	recall	f1-score	support
False	0.95	0.82	0.88	29238
True	0.31	0.65	0.42	3712
micro avg	0.80	0.80	0.80	32950
macro avg	0.63	0.74	0.65	32950
weighted avg	0.88	0.80	0.83	32950

	Predicted Negative	Predicted Positive
Actual Negative	23971	5267
Actual Positive	1294	2418

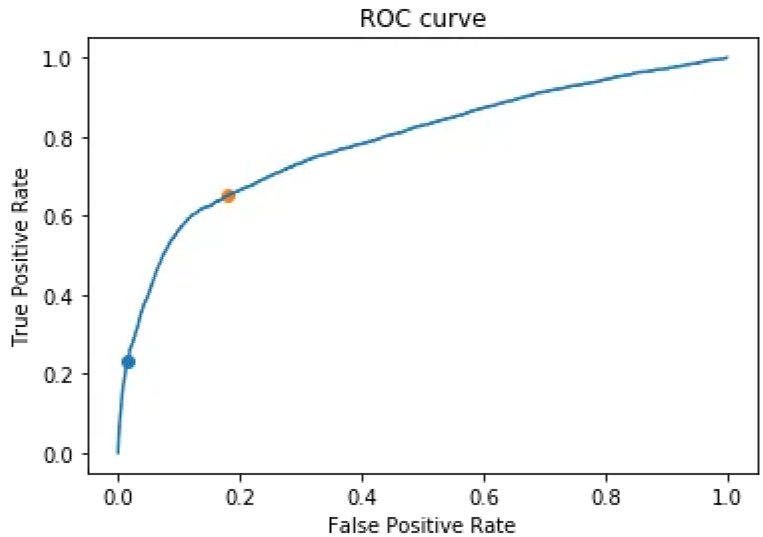
roc_auc_score from sklearn:

```
sklearn.metrics import roc_auc_score
roc_auc_score(y_val, y_pred)
```

The roc_auc_score always runs from 0 to 1, and is sorting predictive possibilities. 0.5 is the baseline for random guessing, so you want to always get above 0.5.

Generating an ROC curve:

ROC stands for curves receiver or operating characteristic curve. It illustrates in a binary classifier system the discrimination threshold created by plotting the true positive rate vs false positive rate.



The perfect point is at the left top, where there are a lot of true positives and less false positives.

```
from sklearn.metrics import roc_auc_score, roc_curve
fpr, tpr, thresholds = roc_curve(y_train, y_pred_proba)
plt.plot(fpr, tpr)
plt.title('ROC curve')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
print('Area under the Receiver Operating Characteristic curve:',
      roc_auc_score(y_train, y_pred_proba))

# When threshold = 0.5
false_positives = 452
true_positives = 860
false_positive_rate = false_positives/actual_negative
true_positive_rate = true_positives/actual_positive
plt.scatter(false_positive_rate, true_positive_rate)

# When threshold = 0.1
false_positives = 5267
true_positives = 2418
false_positive_rate = false_positives/actual_negative
true_positive_rate = true_positives/actual_positive
plt.scatter(false_positive_rate, true_positive_rate);
```

Essentially the above graph plots the following dataframe of false positive rates and true positive rates:

```
pd.DataFrame(dict(fpr=fpr, tpr=tpr, thresholds=thresholds))
```

1	0.000000	0.000269	0.903011
2	0.000000	0.004849	0.880439
3	0.000034	0.004849	0.879162
4	0.000034	0.008890	0.872156
5	0.000103	0.008890	0.871648
6	0.000103	0.009429	0.870454
7	0.000137	0.009429	0.869912
8	0.000137	0.011584	0.864618
9	0.000171	0.011584	0.864278

```
from sklearn.metrics import roc_auc_score
roc_auc_score(y_val, y_pred)
```

The roc_auc_score always runs from 0 to 1, and is sorting predictive possibilities. 0.5 is the baseline for random guessing, so you want to always get above 0.5.

Class imbalance:

In binary classification using logistic regression, we might not be predicting something that has a 50–50 chance. Sometimes in fraudulent cases, positives occur in a small fraction of cases. How do we check if indeed our dataset exhibits class imbalance?

```
y_train.value_counts(normalize=True)
```

```
Name: fail, dtype: float64
```

As you can see there are a lot more 0's, and the split is not 50–50.

So what do we do if we need to weigh our positives higher?

The class_weight parameter:

class_weight = None means errors are equally weighted, however sometimes mis-classifying one class might be worse.

class_weight = 'balanced' means that instead for each observation to be weighted equally, each class is weighted equally, helps with auc_roc score.

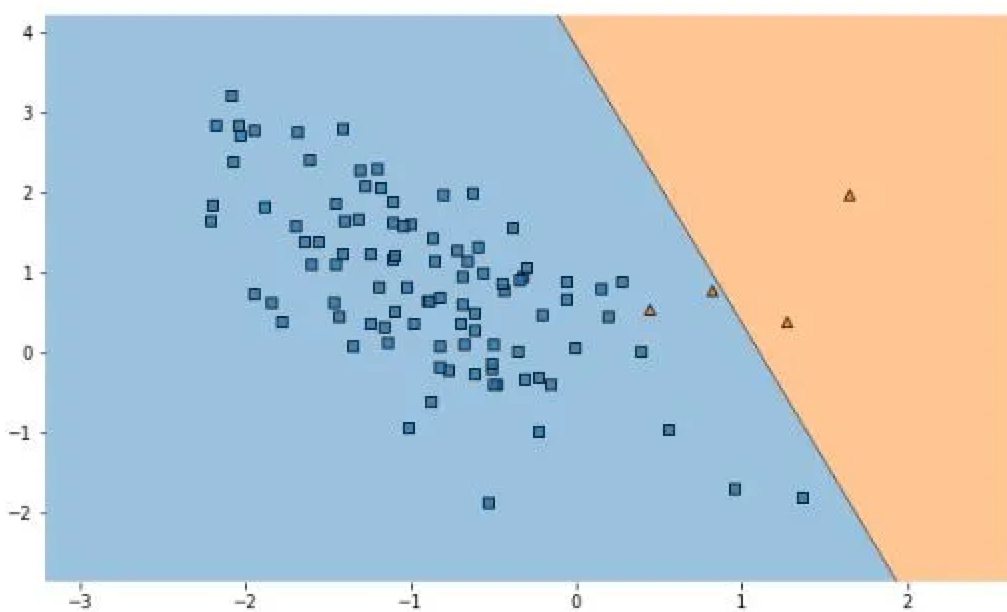
With a class_weight = {0:1, 1:10}, the second value is weighted 10 times greater than the first.

The predicted probablities need to be passed in for roc_auc_score, comparing ground truth to predicted probabilities.

```
class_weight = None

model = LogisticRegression(solver='lbfgs', class_weight=class_weight)
model.fit(X_train, y_train)


# 4. Evaluate model

y_pred = model.predict(X_val)
print(classification_report(y_val, y_pred))
print('accuracy', accuracy_score(y_val, y_pred))
display(pd.DataFrame(
    confusion_matrix(y_val, y_pred),
    columns=['Predicted Negative', 'Predicted Positive'],
    index=['Actual Negative', 'Actual Positive']))


# 5. Visualize decision regions

plt.figure(figsize=(10, 6))
plot_decision_regions(X_val, y_val, model, legend=0);
```

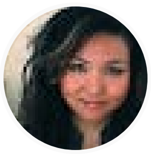
```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        96
           1       1.00      0.50      0.67         4

   micro avg       0.98      0.98      0.98       100
   macro avg       0.99      0.75      0.83       100
weighted avg       0.98      0.98      0.98       100

accuracy 0.98
```

	Predicted Negative	Predicted Positive
Actual Negative	96	0
Actual Positive	2	2



Was this helpful? Here are [24 more evaluation metrics]() to consider when working with binary classification.

Precision Recall Imbalanced Class Confusion Matrix Roc Curve



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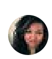

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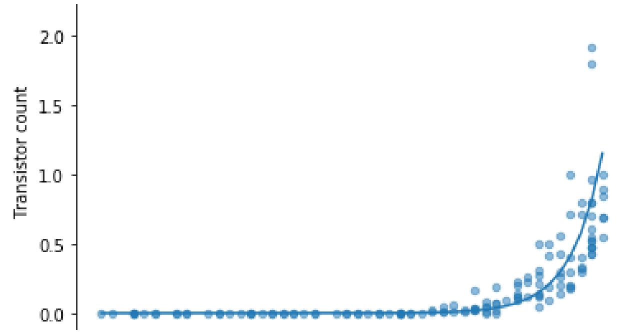

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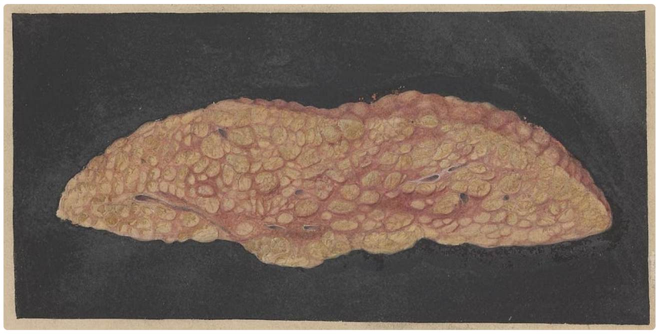

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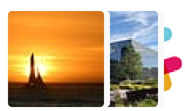

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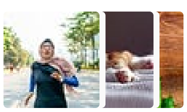

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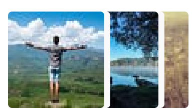

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